Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

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Worked Solutions **Advanced Theorems and Applications**

Q001 **Written**

Find the maximum and minimum of $y = sin^2\theta + cos\theta + 1, 0le\theta < 2pi$, and the corresponding θ.

Solution

1. Put $t = cos\theta$: $y = -t^2 + t + 2 = -(t - frac12)^2 + frac94$.
2. Check the vertex and the endpoints $t = -1, 1$, then return to θ.

Final answer

$\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \pi$

Q002 **Written**

Find the maximum and minimum of $y = cos^2\theta - sin\theta, 0le\theta < 2pi$, and the corresponding θ.

Solution

1. Put $t = sin\theta$: $y = 1 - t^2 - t = -(t + frac12)^2 + frac54$.
2. The endpoint $t = 1$ gives the minimum -1.

Final answer

$\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}$

Worked Solutions **Advanced Theorems and Applications**

Q003 Written

Solve the four given maximum/minimum problems on $0le\theta < 2pi$, including $y = \sin^2\theta + \sin\theta - 1$, $y = \cos^2\theta - \cos\theta + 1$, $y = 2\sin\theta + \cos^2\theta - 1$, and $y = -2\sin^2\theta - 2\cos\theta + 1$. Include the given challenge $y = \cos^2\theta - \sin\theta$.

Solution

1. Choose the single trig variable that makes the expression quadratic.
2. The vertex is valid only when it lies in $[-1, 1]$; convert every extremal t back to the corresponding angles.

Final answer

See the numbered solution: substitute $t = \sin\theta$ or $t = \cos\theta$, complete the square, then test $t = \pm 1$.

Worked Solutions **Advanced Theorems and Applications**

Q004 **Written**

Use addition theorems to find $\sin 75°$ and $\cos(\pi/12)$.

Solution

1. Rewrite $75° = 45° + 30°$ and $pi/12 = 45° - 30°$.
2. Apply the sine-sum and cosine-difference identities.

Final answer

$\sin 75° = \cos \pi/12 = \dfrac{\sqrt{6}+\sqrt{2}}{4}$

Q005 **Written**

Given $\sin\alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos\beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha+\beta)$ and $\cos(\alpha-\beta)$.

Solution

1. The quadrant signs give $\cos\alpha = -1/2$ and $\sin\beta = -4/5$.
2. Substitute in the addition identities without a calculator.

Final answer

$\sin\alpha+\beta = \dfrac{4-3\sqrt{3}}{10}; \quad \cos\alpha-\beta = \dfrac{3-4\sqrt{3}}{10}$

Worked Solutions   **Advanced Theorems and Applications**

Q006 **Written**

Complete the given addition-theorem exercises $exactvalues, quadrant - controlled\alpha, \beta, andthethree - anglechallenge(\sin\alpha = 0.6, \cos\beta = 0.8, \tan\gamma = 0.75)$**.**

Solution

1. Convert radians to degrees only when it clarifies the reference angle.
2. Recover the missing sine/cosine value from the quadrant before applying the sum or difference identity.

Final answer

See the numbered solution; the three-angle challenge gives $\sin(\alpha+\beta+\gamma) = \frac{117}{125}$.

Q007 **Written**

Find $\tan 105°$**, then find the angle between** $y = 2x + 1$ **and** $y = -3x + 2$**.**

Solution

1. Use $105° = 45° + 60°$ and the tangent addition theorem.
2. For the lines, use $m_1 = 2, m_2 = -3$ in $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$.

Final answer

$\tan 105° = -2 + \sqrt{3}; \quad \theta = \frac{\pi}{4}$

Worked Solutions **Advanced Theorems and Applications**

Q008 **Written**

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x+2y-3 = 0$ with $-5x+3y+1 = 0$.

Solution

1. Convert each line to slope-intercept form.
2. Use the acute angle convention $0 < \theta < \pi/2$, replacing θ_1 by $pi - \theta_1$ when necessary.

Final answer

The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Worked Solutions **Advanced Theorems and Applications**

Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan\theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y-axis.

Solution

1. Solve the two linear equations produced by the tangent formula for the unknown slope.
2. Use the y-intercepts and the line intersection to obtain each triangle area.

Final answer

See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Worked Solutions **Advanced Theorems and Applications**

Q010 **Written**

Given $\sin\alpha = 1/3$ **with α in QII, find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$**.**

Solution

1. The quadrant gives $\cos\alpha = -2\sqrt{2}/3$.
2. Apply $\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$, then divide.

Final answer

$$\sin 2\alpha = -\frac{4\sqrt{2}}{9}, \quad \cos 2\alpha = \frac{7}{9}, \quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$$

Q011 **Written**

Given $\sin\alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$**, find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$**.**

Solution

1. The interval places α in QIII, so $\cos\alpha = -\sqrt{5}/3$.
2. Substitute into the three double-angle formulas.

Final answer

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}, \quad \cos 2\alpha = \frac{1}{9}, \quad \tan 2\alpha = 4\sqrt{5}$$

Worked Solutions **Advanced Theorems and Applications**

Q012 **Written**

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos\theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Solution

1. Determine the missing trig value from the interval before using the identities.
2. For 3θ, use $4cos^3\theta - 3cos\theta$; for 2θ, use the tangent double-angle identity.

Final answer

For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 **Written**

Given $\cos\alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

Solution

1. Use the half-angle formulas and choose signs from $0 < a/2 < pi/2$.
2. For (5pi/8), use $cos(\pi - 3\pi/8)$.

Final answer

$$\sin\frac{\alpha}{2} = \frac{1}{\sqrt{5}}, \quad \cos\frac{\alpha}{2} = \frac{2}{\sqrt{5}}, \quad \tan\frac{\alpha}{2} = \frac{1}{2}, \quad \cos\frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

Worked Solutions **Advanced Theorems and Applications**

Q014 **Written**

Given $\cos\alpha = -1/3, 0 < \alpha < \pi$, **find** $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Solution

1. Since α/2 is in QI, both half-angle roots are positive.
2. Use $3pi/8 = \pi/2 - \pi/8$ or the half-angle identity.

Final answer

$$\sin\frac{\alpha}{2} = \sqrt{\frac{2}{3}}, \quad \cos\frac{\alpha}{2} = \frac{1}{\sqrt{3}}, \quad \tan\frac{\alpha}{2} = \sqrt{2}, \quad \sin\frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$$

Q015 **Written**

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi$.

Solution

1. Halve the interval first; its quadrant controls every sign.
2. Apply the half-angle formula again for (A/4) after finding (cos(A/2)).

Final answer

Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}}-\sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Worked Solutions **Advanced Theorems and Applications**

Q016 **Written**

Solve on $0 \le \theta < 2\pi$: a $\sin 2\theta + \sin\theta = 0$, b $\cos 2\theta - 7\cos\theta + 4 \ge 0$.

Solution

1. Use $\sin 2\theta = 2\sin\theta\cos\theta$ and factor.
2. Use $\cos 2\theta = 2\cos^2\theta - 1$, solve the quadratic in $c = \cos\theta$, and map the interval.

Final answer

$a\ \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}; \quad b\ \frac{\pi}{3} \le \theta \le \frac{5\pi}{3}$

Q017 **Written**

Solve the given equation $\sin 2\theta + \cos\theta = 0$ on $0 \le \theta < 2\pi$.

Solution

1. Factor $cos\theta(2sin\theta + 1) = 0$.
2. List the two cosine zeros and the two angles with $sin\theta = -1/2$.

Final answer

$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

Worked Solutions **Advanced Theorems and Applications**

Q018 Written

Solve all given double-angle equations and inequalities on $0 \le \theta < 2\pi$, including the challenge $\cos 2\theta - 7\cos\theta + 4 \ge 0$.

Solution

1. Choose the identity that leaves one trig function only.
2. For inequalities, use a sign chart and include or exclude boundary points exactly as stated.

Final answer

See the numbered solution: replace double angles, solve in $s = \sin\theta$ or $c = \cos\theta$, and translate the accepted ranges.

Q019 Written

Rewrite $\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta + \alpha)$, $r > 0, -\pi < \alpha < \pi$.

Solution

1. Plot $P(1, -\sqrt{3})$, find $r = OP$, then read the angle α.

Final answer

$$r = 2, \quad \alpha = -\frac{\pi}{3}, \quad \sin\theta - \sqrt{3}\cos\theta = 2\sin\theta - \frac{\pi}{3}$$

Worked Solutions **Advanced Theorems and Applications**

Q020 Written

Rewrite the given expressions $\sin\theta-\cos\theta$, $\sin\theta+\cos\theta$, $3\sin\theta-\sqrt{3}\cos\theta$, and $3\sin\theta-\cos\theta$ in harmonic-addition form.

Solution

1. For $a\sin\theta+b\cos\theta$, use $r=\sqrt{a^2+b^2}$, $\cos\alpha=a/r$, $\sin\alpha=b/r$.

Final answer

$\sqrt{2}\sin\theta-\frac{\pi}{4}$, $\sqrt{2}\sin\theta+\frac{\pi}{4}$, $2\sqrt{3}\sin\theta-\frac{\pi}{6}$, $\sqrt{10}\sin\theta-\arctan\frac{1}{3}$

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r\sin(\theta+\alpha)$, including $2\sin\theta+2\cos\theta$ and the challenge $\sin\theta-\sqrt{3}\cos\theta$.

Solution

1. Keep the sign of the cosine coefficient when locating P(a,b).
2. Substitute the obtained (r,a) and expand once to verify both coefficients.

Final answer

See the numbered solution; every result has $r=\sqrt{a^2+b^2}>0$ and α chosen in $(-\pi,\pi)$.

Worked Solutions **Advanced Theorems and Applications**

Q022 **Written**

Solve on $0 \le \theta < 2\pi$: $a \sin\theta - \cos\theta = 1$, $b \sin\theta + \cos\theta < 3/2$.

Solution

1. Synthesize each left-hand side and account for the shifted angle range.
2. The maximum of $\sin\theta + \cos\theta$ is $\sqrt{2} < 3/2$, so the second inequality is always true.

Final answer

$a\ \theta = \frac{\pi}{2}, \pi; \quad b\ 0 \le \theta < 2\pi$

Q023 **Written**

Solve $\sin\theta \le \cos\theta$ on $0 \le \theta < 2\pi$.

Solution

1. Rewrite as $\sqrt{2}\sin(\theta - \pi/4) \le 0$.
2. Read the accepted half-turns and retain the equality endpoints.

Final answer

$0 \le \theta \le \frac{\pi}{4} \quad \text{or} \quad \frac{5\pi}{4} \le \theta < 2\pi$

Worked Solutions **Advanced Theorems and Applications**

Q024 Written

Solve all given harmonic-addition equations and inequalities on $0 \le \theta < 2\pi$, including $\sin\theta + \cos\theta + 1 = 0$ and $\sin\theta - \cos\theta + 3/2 > 0$.

Solution

1. After synthesis set $t = \theta + \alpha$ and write the exact t-range.
2. Use one full revolution in t, then subtract α and intersect with the original domain.

Final answer

See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 Written

Find the maximum and minimum of $y = 2\sin(\theta - \pi/3)$, and of $y = 2\cos 2\theta + 4\cos\theta + 1$, $0 \le \theta < 2\pi$.

Solution

1. Use $-1 \le \sin(t) \le 1$ for the first function.
2. For the second, set $t = \cos\theta$ and complete the square after the double-angle substitution.

Final answer

First: max $= 2$, min $= -2$. Second: max $= 7$ at $\theta = 0$; min $= -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions **Advanced Theorems and Applications**

Q026 **Written**

Find the maximum and minimum of the given functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

Solution

1. Reduce double angles or $sin^2\theta$ to one variable.
2. Check the vertex and both endpoints of the allowed interval.

Final answer

For the first, $\max = 1, \min = -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 **Written**

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Solution

1. Use the same transformation rules as for equations and inequalities.
2. Never report a vertex outside $[-1, 1]$; compare it with both endpoints.

Final answer

See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Worked Solutions **Advanced Theorems and Applications**

Q028 MCQ

The substitution for $\sin^2\theta + \cos\theta + 1$ should be:

Solution

1. Use $\sin^2\theta = 1 - t^2$.

Correct option: B

Final answer

$t = cos$ theta

Q029 MCQ

The maximum of $\sin^2\theta + \cos\theta + 1$ is:

Solution

1. Complete the square.

Correct option: C

Final answer

9/4

Q030 MCQ

An extremal vertex is valid only if t lies in:

Solution

1. Trig sine and cosine values are bounded.

Correct option: B

Final answer

$[-1, 1]$

Q031 MCQ

The identity for sin(a+b) is:

Solution

1. Use the addition theorem.

Correct option: A

Final answer

sin a cos b + cos a sin b

Q032 MCQ

sin 75 degrees equals:

Solution

1. Rewrite 75 as 45+30.

Correct option: B

Final answer

$sqrt6 + \sqrt{2}/4$

Q033 MCQ

A missing trig value is selected using:

Solution

1. The quadrant fixes the sign.

Correct option: A

Final answer

the quadrant

Worked Solutions **Advanced Theorems and Applications**

Q034 MCQ

The slope of y=mx+k is:

Solution

1. The coefficient of x is the slope.

Correct option: B

Final answer

m

Q035 MCQ

The angle formula uses denominator:

Solution

1. Use the tangent difference theorem.

Correct option: A

Final answer

$1 + m1m2$

Q036 MCQ

For an acute angle, if tan theta1 is negative we use:

Solution

1. Choose the acute representative.

Correct option: B

Final answer

pi-theta1

Q037 MCQ

sin(2a) equals:

Solution

1. Double-angle identity.

Correct option: B

Final answer

2 sin a cos a

Worked Solutions **Advanced Theorems and Applications**

Q038 MCQ

cos(2a) can be written as:

Solution

1. Use the standard formula.

Correct option: A

Final answer

$\cos^2 a - sin^2$ a

Q039 MCQ

For theta in QIV, sin theta is:

Solution

1. Read the quadrant.

Correct option: B

Final answer

negative

Worked Solutions **Advanced Theorems and Applications**

Q040 MCQ

The half-angle formula for sin(a/2) uses:

Solution

1. Choose the sign from a/2.

Correct option: A

Final answer

sqrt((1-cos a)/2)

Q041 MCQ

The sign of a half-angle root is chosen from:

Solution

1. Halving changes the interval.

Correct option: A

Final answer

the original quadrant interval

Worked Solutions **Advanced Theorems and Applications**

Q042 MCQ

If cos a=-1/3 and 0<a<pi, cos(a/2) is:

Solution

1. a/2 is in QI.

Correct option: B

Final answer

positive

Q043 MCQ

For double-angle equations, first rewrite in:

Solution

1. Use a double-angle identity.

Correct option: A

Final answer

one trig function

Worked Solutions **Advanced Theorems and Applications**

Q044 MCQ

The factorisation of sin(2 theta)+sin theta is:

Solution

1. Factor after substituting sin(2 theta).

Correct option: A

Final answer

sin theta(2cos theta+1)

Q045 MCQ

A tangent asymptote is always:

Solution

1. Tangent is undefined there.

Correct option: B

Final answer

excluded

Worked Solutions **Advanced Theorems and Applications**

Q046 MCQ

For a sin theta+b cos theta, r is:

Solution

1. It is the length OP.

Correct option: B

Final answer

sqrt$a^2 + b^2$

Q047 MCQ

The harmonic target form is:

Solution

1. Match the coefficients.

Correct option: A

Final answer

r sin(theta+a)

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Q048 MCQ

The angle a is located from point:

Solution

1. The coefficient point determines direction.

Correct option: A

Final answer

$P(a, b)$

Q049 MCQ

After synthesis, t is usually:

Solution

1. Track the shifted range.

Correct option: A

Final answer

theta+a

Worked Solutions **Advanced Theorems and Applications**

Q050 MCQ

If max(sin theta+cos theta)=sqrt2, then sin theta+cos theta<3/2 is:

Solution

1. sqrt2<3/2.

Correct option: B

Final answer

always true

Q051 MCQ

In a strict inequality, equality endpoints are:

Solution

1. Use open endpoints.

Correct option: B

Final answer

excluded

Q052 MCQ

The range of 2sin(theta-pi/3) is:

Solution

1. Multiply the sine range by 2.

Correct option: B

Final answer

$[-2, 2]$

Q053 MCQ

For a double-angle maximum problem, use:

Solution

1. Reduce to $t = sin$ theta or cos theta.

Correct option: A

Final answer

the double-angle formula then a bounded variable

Worked Solutions **Advanced Theorems and Applications**

Q054 MCQ

A vertex outside [-1,1] is:

Solution

1. It is not attainable by sine/cosine.

Correct option: B

Final answer

rejected

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2\sin(\theta - \pi/3)$ is:

Solution

1. Use the sine range.

Correct option: D

Final answer

2

Worked Solutions **Advanced Theorems and Applications**

Quiz MCQ 2 **Concept Quiz · MCQ**

The addition-theorem value $\sin 75°$ is:

Solution

1. Use 45 degrees plus 30 degrees.

Correct option: A

Final answer

$\sqrt{6} + \sqrt{2}/4$

Quiz MCQ 3 **Concept Quiz · MCQ**

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Solution

1. Complete the square.

Correct option: A

Final answer

-2

Worked Solutions **Advanced Theorems and Applications**

Quiz MCQ 4 Concept Quiz · MCQ

The angle between slopes 2 and -3 is:

Solution

1. The tangent magnitude is 1.

Correct option: B

Final answer

$\pi/4$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$ on $0 \le \theta < 2\pi$.

Solution

1. Let $t = \cos\theta$ and complete the square.

Final answer

$\max = \frac{9}{4}, \min = 0$

Worked Solutions **Advanced Theorems and Applications**

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$ when $\sin\alpha = 1/3$ and α is in QII.

Solution

1. Use the quadrant to select $\cos\alpha$.

Final answer

$-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta+\alpha)$.

Solution

1. Find r and the coefficient angle from P(3,-sqrt3).

Final answer

$2\sqrt{3}\sin\theta - \frac{\pi}{6}$

Worked Solutions **Advanced Theorems and Applications**

Quiz WRITTEN 8 Concept Quiz · Written

Solve $\sin\theta \leq \cos\theta$ on $0 \leq \theta < 2\pi$.

Solution

1. Synthesize to $\sqrt{2}\sin(\theta - \pi/4) \leq 0$.

Final answer

$[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Answer key

Use the question number shown on each page.

Q001 $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\quad \min y = 0$ at $\theta = \pi$

Q002 $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\quad \min y = -1$ at $\theta = \frac{\pi}{2}$

Q003 See the numbered solution: substitute $t = \sin\theta$ or $t = \cos\theta$, complete the square, then test $t = \pm 1$.

Q004 $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6}+\sqrt{2}}{4}$

Q005 $\sin \alpha + \beta = \frac{4-3\sqrt{3}}{10}$; $\quad \cos \alpha - \beta = \frac{3-4\sqrt{3}}{10}$

Q006 See the numbered solution; the three-angle challenge gives $\sin(\alpha+\beta+\gamma) = \frac{117}{125}$.

Q007 $\tan 105^\circ = -2+\sqrt{3}$; $\quad \theta = \frac{\pi}{4}$

Q008 The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Q009 See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas 49/165 and 25/3.

Q010 $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\quad \cos 2\alpha = \frac{7}{9}$, $\quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$

Q011 $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\quad \cos 2\alpha = \frac{1}{9}$, $\quad \tan 2\alpha = 4\sqrt{5}$

Q012 For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 $\sin\frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\quad \cos\frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\quad \tan\frac{\alpha}{2} = \frac{1}{2}$, $\quad \cos\frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$

Q014 $\sin\frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\quad \cos\frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\quad \tan\frac{\alpha}{2} = \sqrt{2}$, $\quad \sin\frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$

Q015 Use the signed half-angle formulas; for the challenge,

$$\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6},$$

$$\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}.$$

Q016 a $\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $\quad$ b $\frac{\pi}{3} \le \theta \le \frac{5\pi}{3}$

Q017 $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

Q018 See the numbered solution: replace double angles, solve in $s = \sin\theta$ or $c = \cos\theta$, and translate the accepted ranges.

Q019 $r = 2$, $\quad \alpha = -\frac{\pi}{3}$, $\quad \sin\theta - \sqrt{3}\cos\theta = 2\sin\theta - \frac{\pi}{3}$

Q020 $\sqrt{2}\sin\theta - \frac{\pi}{4}$, $\sqrt{2}\sin\theta + \frac{\pi}{4}$, $2\sqrt{3}\sin\theta - \frac{\pi}{6}$, $\sqrt{10}\sin\theta - \arctan\frac{1}{3}$

Q021 See the numbered solution; every result has $r = \sqrt{a^2+b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Q022 a $\theta = \frac{\pi}{2}, \pi$; $\quad$ b $0 \le \theta < 2\pi$

Q023 $0 \le \theta \le \frac{\pi}{4}$ $\quad$ or $\quad \frac{5\pi}{4} \le \theta < 2\pi$

Q024 See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 First: max $= 2$, min $= -2$. Second: max $= 7$ at $\theta = 0$; min $= -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Q026 For the first, max $= 1$, min $= -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Q028 B

Q029 C

Q030 B

Q031 A

Q032 B

Q033 A

Q034 B

Q035 A

Q036 B

Q037 B

Q038 A

Q039 B

Q040 A

Q041 A

Q042 B

Q043 A

Q044 A

Q045 B

Q046 B

Q047 A

Q048 A

Q049 A

Q050 B

Q051 B

Q052 B

Q053 A

Q054 B

Quiz MCQ 1 D

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 max $= \frac{9}{4}$, min $= 0$

Quiz WRITTEN 6 $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$

Quiz WRITTEN 7 $2\sqrt{3}\sin\theta - \frac{\pi}{6}$

Quiz WRITTEN 8 $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$